

Byeongjee Kang

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Research Interests

My research is on language and compiler support for automatically generating safe and efficient programs with provable guarantees. I combine static analysis with compiler infrastructure built on LLVM and MLIR. My recent interests are in resource-constrained platforms, where programs require checkpointing or migration to safely execute under strict energy and memory budgets. I am also broadly interested in parallel programming, formal verification, and safe systems programming in Rust.

Technical Skills

Expertise: Compiler engineering, static analysis, information flow analysis, embedded systems, formal verification

Languages & Tools: Rust, C++, Python, SML, Maude, Futhark, WebAssembly, LLVM, MLIR, rustc (MIR)

Education

PhD	Carnegie Mellon University , Software Engineering Advised by Prof. Limin Jia	Pittsburgh, USA Aug 2023 – May 2028 (expected)
BS	Pohang University of Science and Technology (POSTECH) , Computer Science and Engineering & Mathematics (double major) - GPA: 4.26/4.3 (Ranked 1st among all graduating students in class of 2023) - Entered with presidential scholarship offered to one outstanding high school student in the year, and graduated with the highest GPA - Early graduation by one-year (leave of absence for military duty during Jan 2020 – Dec 2021)	Pohang, Korea Feb 2018 – Feb 2023

Work Experiences

Microsoft Research , Research intern at Data Systems Group mentored by Kaushik Rajan Built auto-vectorizing compiler optimization passes using MLIR and Polygeist, vectorizing loops with control-flow dependencies that are not supported by existing loop vectorizers	Redmond, WA, USA May 2025 – Aug 2025
Kodebox , Software engineer as military service alternative - Developed the consensus engine of CodeChain Foundry, an open-source blockchain engine with a composable module system, implementing BLS signature aggregation for cryptographic vote verification and the interface between modules and the consensus engine - Built Zuzu, a web-based cap table management software, as a full-stack engineer	Seoul, Korea Jan 2020 – Dec 2021

Publications

Bao: Automatic Region Placement and Memory Allocation for Intermittent Computing

LLVM-based compiler that optimizes checkpoint placement and memory allocation for intermittently powered devices by solving a mixed-integer linear program

Byeongjee Kang, Jan Hoffmann, Limin Jia, Brandon Lucia, Feras Saad

(International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS) 2027 (to appear)) [\[code\]](#)

WAMI: Compilation to WebAssembly through MLIR without Losing Abstraction

Aug 2026

MLIR-based compiler from high-level languages to WebAssembly that takes advantage of high-level Wasm features such as stack switching

Byeongjee Kang, Harsh Desai, Limin Jia, Brandon Lucia

[10.1145/3828170.3828172](https://doi.org/10.1145/3828170.3828172) (International Conference on Managed Programming Languages and Runtimes (MPLR) 2026) [\[code\]](#)

Fray: An Efficient General-Purpose Concurrency Testing Platform for the JVM

Oct 2025

Concurrency testing platform for Java. Formalized a subset of Java concurrency semantics and proved that scheduling only at synchronization points is complete for data-race-free programs

Ao Li, **Byeongjee Kang**, Vasudev Vikram, Isabella Laybourn, Samvid Dharanikota, Shrey Tiwari, Rohan Padhye

[10.1145/3764119](https://doi.org/10.1145/3764119) (Object-oriented Programming, Systems, Languages, and Applications (OOPSLA) 2025) [\[code\]](#)

GraFeyn: Efficient Parallel Sparse Simulation of Quantum Circuits

Sept 2024

Sam Westrick, Pengyu Liu, **Byeongjee Kang**, Colin McDonald, Mike Rainey, Mingkuan Xu, Jatin Arora, Yongshan Ding, Umut Acar

ieeexplore.ieee.org/document/10821338 (IEEE International Conference on Quantum Computing and Engineering (QCE) 2024 (Best Paper))

Narrowing and Heuristic Search for Symbolic Reachability Analysis of Concurrent Object-Oriented Systems

July 2024

Refutation-complete symbolic model checking of distributed systems in Maude using narrowing with heuristic search

Byeongjee Kang, Kyungmin Bae

[10.1016/j.scico.2024.103097](https://doi.org/10.1016/j.scico.2024.103097) (Science of Computer Programming 2024) [\[code\]](#)

Symbolic Reachability Analysis of Distributed Systems using Narrowing and Heuristic Search

Dec 2022

Byeongjee Kang, Kyungmin Bae

[dl.acm.org/doi/10.1145/3563822.3568017](https://doi.org/10.1145/3563822.3568017) (International Workshop on Formal Techniques for Safety-Critical Systems (FTSCS) 2022) [\[code\]](#)

Software

- **Bao**: LLVM toolchain that instruments programs for batteryless devices by solving a mixed-integer linear program
- **WAMI**: MLIR-based compilation pipeline for WebAssembly
- **OCaMI** (work in progress): Compiler from a functional subset of OCaml to WebAssembly through WAMI
- **rustlane** (work in progress): ISPC-style SPMD programming for Rust, lowering scalar control flow to masked SIMD
- **Fray**: Controlled concurrency testing framework for the JVM
- **heuristic-narrowing-search**: Maude implementation of narrowing-based symbolic reachability analysis
- **Simperby**: BFT blockchain engine for decentralized autonomous organizations (DAOs). Designed **Vetomint**, its consensus protocol, which extends Tendermint with leader veto and early termination
- **CodeChain Foundry**: Open-source blockchain engine based on composable module system

Ongoing Research Projects

- **Language-Level Program Migration**: Compiler and runtime support for migrating live Rust programs under strict time and energy budgets by erasing reinitializable state at the source and rebuilding it at the destination
- **Probabilistic Energy Estimation**: Probabilistic modeling and analysis tools for estimating energy consumption of programs on resource-constrained devices
- **Cumulative Memory Analysis**: Type-based automatic amortized resource analysis that infers symbolic bounds on memory usage integrated over execution time

Honors and Scholarships

ILJU Fellowship (ILJU Academy and Culture Foundation)

Aug 2023 – July 2028

- Received total of USD 120,000 during graduate studies

Global Leadership Program (CSE Department, POSTECH)

Sept 2019 – Feb 2023

- Scholarship and research support program for selected students in the CS department

Domun Scholarship (POSTECH)

Feb 2019

- Scholarship for freshmen with the highest GPA

Program for Highly Dedicated Students (POSTECH)

Feb 2018 – Feb 2023

- Presidential scholarship and supporting program in POSTECH offered to one outstanding freshman each year

Presidential Science Scholarship (Ministry of Education of Korea)

Feb 2018 – Feb 2023

- National scholarship offered to top freshmen students in the field of science and technology

Talks

- An MLIR Dialect for WebAssembly, WebAssembly Workshop 2025, [[recording](#)]

Service

- Artifact Evaluation Committee, POPL 2027